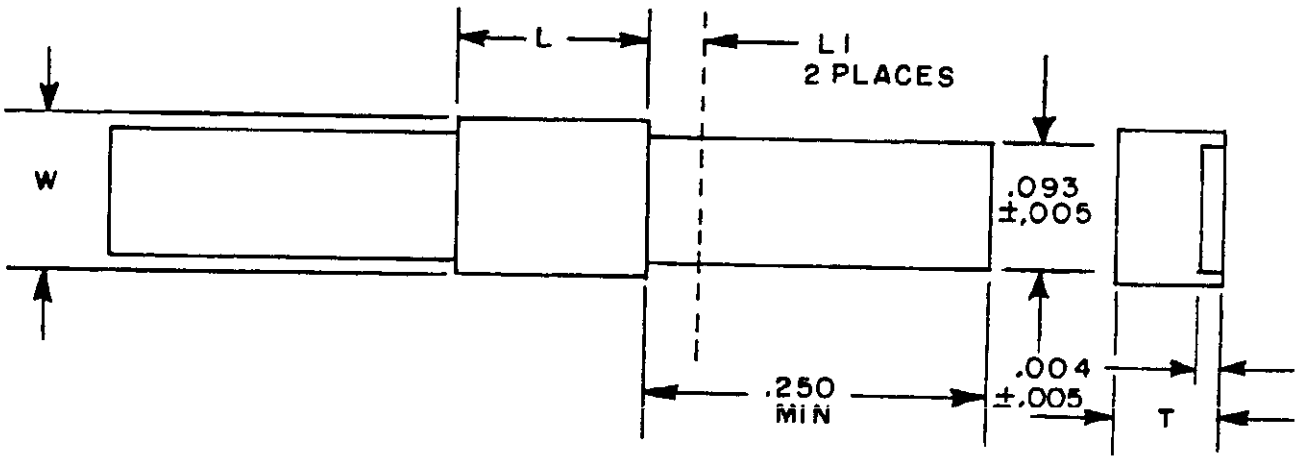
MILITARY SPECIFICATION SHEET

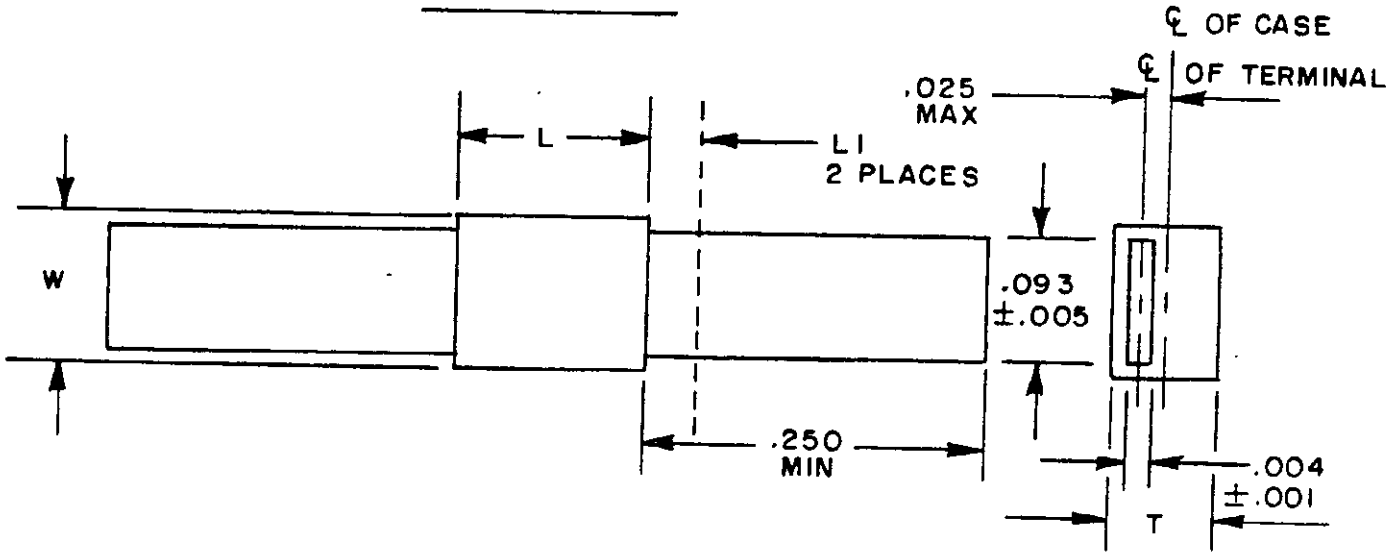
CAPACITORS, CHIP, MULTIPLE LAYER, FIXED, UNENCAPSULATED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY STYLES CDR21, CDR22, CDR23, CDR24, AND CDR25 (HIGH FREQUENCY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-55681.



STYLE CDR21

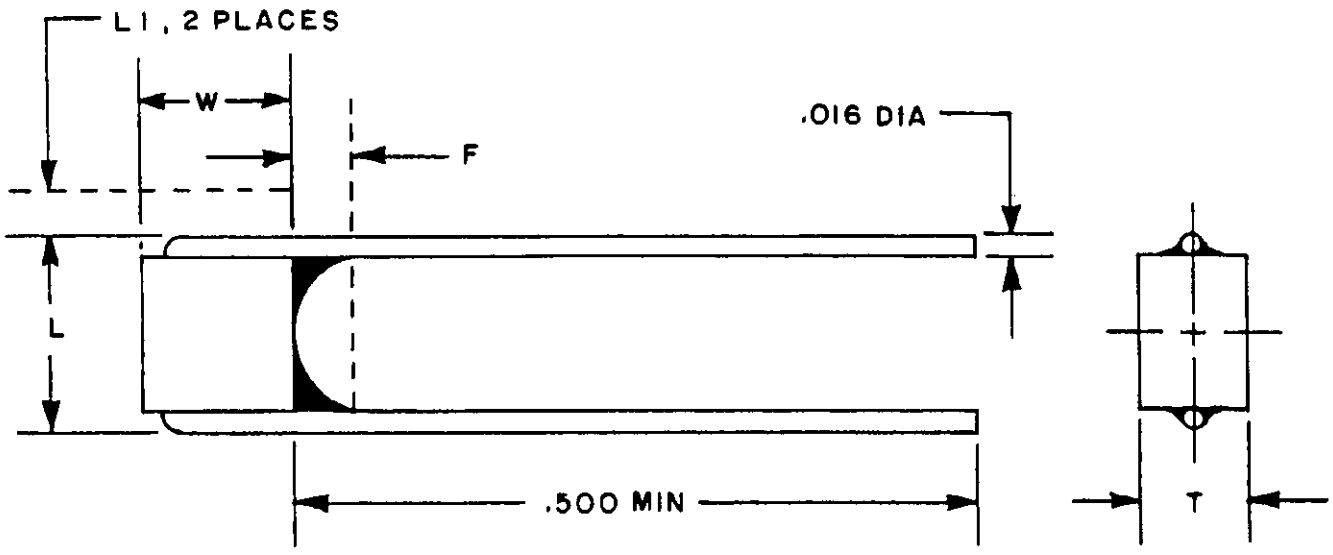


STYLE CDR22

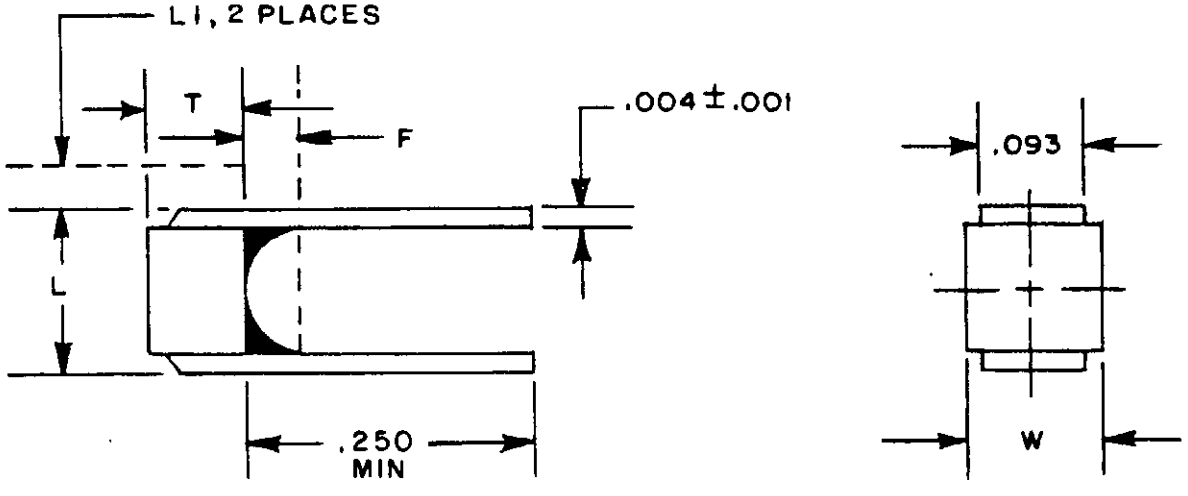
FIGURE 1. Dimensions and configuration.

FSC 5910

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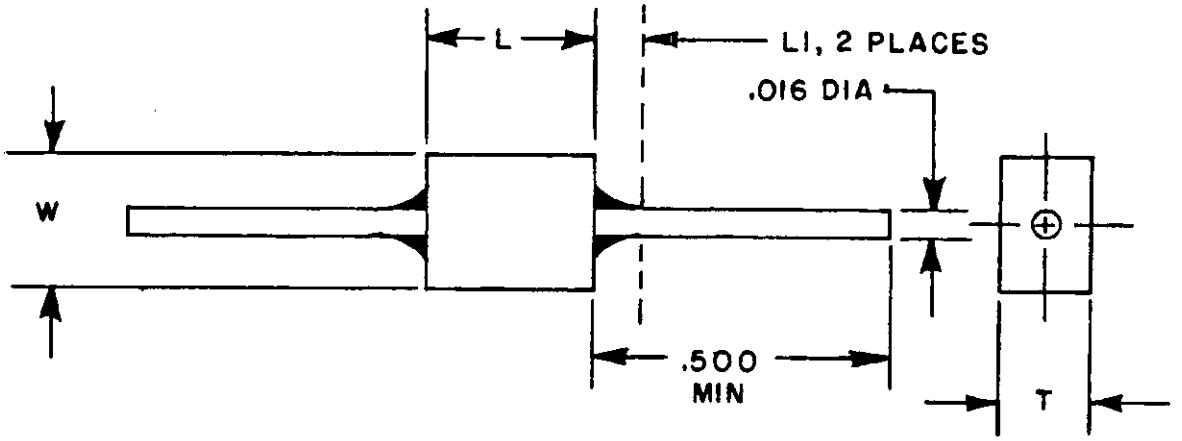
STYLE CDR23



STYLE CDR24

FIGURE 1. Dimensions and configuration - Continued.

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STYLE CDR25



STYLE CDR21



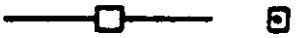
STYLE CDR22



STYLE CDR23



STYLE CDR24



STYLE CDR25

APPROXIMATE SIZE

Style	Dimensions L	W	T Min	T Max	L1 Max 1/	F Max 2/
CDR21	.135 ±.015	.110 ±.015	.030	.100	.020	----
CDR22	.135 ±.015	.110 ±.015	.030	.100	.020	----
CDR23	.135 ±.015	.110 ±.015	.030	.100	.020	.020
CDR24	.135 ±.015	.110 ±.015	.030	.100	.020	.020
CDR25	.135 ±.015	.110 ±.015	.030	.100	.020	----

1/ Solder build up over each lead.
2/ Solder fill.

INCHES	MM
.001	0.03
.004	0.10
.005	0.13
.015	0.38
.016	0.41
.020	0.51
.025	0.64
.030	0.76
.093	2.36
.100	2.54
.110	2.79
.135	3.43
.250	6.35
.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration - Continued.

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TABLE I. Styles CDR21, CDR22, CDR23, CDR24, and CDR25 capacitor characteristics.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts, dc
CDR2-B-0R1EB--	0.1	B	BG, BP	500
CDR2-B-0R2EB--	0.2	B	"	"
CDR2-B-0R3E---	0.3	B,C	"	"
CDR2-B-0R4E---	0.4	B,C	"	"
CDR2-B-0R5E---	0.5	B,C,D	"	"
CDR2-B-0R6E---	0.6	"	"	"
CDR2-B-0R7E---	0.7	"	"	"
CDR2-B-0R8E---	0.8	"	"	"
CDR2-B-0R9E---	0.9	"	"	"
CDR2-B-1R0E---	1.0	"	"	"
CDR2-B-1R1E---	1.1	"	"	"
CDR2-B-1R2E---	1.2	"	"	"
CDR2-B-1R3E---	1.3	"	"	"
CDR2-B-1R4E---	1.4	"	"	"
CDR2-B-1R5E---	1.5	"	"	"
CDR2-B-1R6E---	1.6	"	"	"
CDR2-B-1R7E---	1.7	"	"	"
CDR2-B-1R8E---	1.8	"	"	"
CDR2-B-1R9E---	1.9	"	"	"
CDR2-B-2R0E---	2.0	"	"	"
CDR2-B-2R1E---	2.1	"	"	"
CDR2-B-2R2E---	2.2	"	"	"
CDR2-B-2R4E---	2.4	"	"	"
CDR2-B-2R7E---	2.7	"	"	"
CDR2-B-3R0E---	3.0	"	"	"
CDR2-B-3R3E---	3.3	"	"	"
CDR2-B-3R6E---	3.6	"	"	"
CDR2-B-3R9E---	3.9	"	"	"
CDR2-B-4R3E---	4.3	"	"	"
CDR2-B-4R7E---	4.7	"	"	"
CDR2-B-5R1E---	5.1	"	"	"
CDR2-B-5R6E---	5.6	"	"	"
CDR2-B-6R2E---	6.2	"	"	"
CDR2-B-6R8E---	6.8	B,C,J,K,M	"	"
CDR2-B-7R5E---	7.5	"	"	"
CDR2-B-8R2E---	8.2	"	"	"
CDR2-B-9R1E---	9.1	"	"	"
CDR2-B-100E---	10	F,G,J,K,M	"	"
CDR2-B-110E---	11	"	"	"
CDR2-B-120E---	12	"	"	"
CDR2-B-130E---	13	"	"	"
CDR2-B-150E---	15	"	"	"
CDR2-B-160E---	16	"	"	"
CDR2-B-180E---	18	"	"	"
CDR2-B-200E---	20	"	"	"
CDR2-B-220E---	22	"	"	"
CDR2-B-240E---	24	"	"	"
CDR2-B-270E---	27	"	"	"
CDR2-B-300E---	30	"	"	"
CDR2-B-330E---	33	"	"	"
CDR2-B-360E---	36	"	"	"
CDR2-B-390E---	39	"	"	"
CDR2-B-430E---	43	"	"	"
CDR2-B-470E---	47	"	"	"
CDR2-B-510E---	51	"	"	"
CDR2-B-560E---	56	"	"	"
CDR2-B-620E---	62	"	"	"
CDR2-B-680E---	68	"	"	"

See footnote at end of table.

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TABLE I. Styles CDR21, CDR22, CDR23, CDR24, and CDR25 capacitor characteristics - Continued.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts, dc
CDR2-B-750E---	75	F,G,J,K,M	BG, BP	500
CDR2-B-820E---	82	"	"	"
CDR2-B-910E---	91	"	"	"
CDR2-B-101E---	100	"	"	"
CDR2-B-111D---	110	"	"	300
CDR2-B-121D---	120	"	"	"
CDR2-B-131D---	130	"	"	"
CDR2-B-151D---	150	"	"	"
CDR2-B-161D---	160	"	"	"
CDR2-B-181D---	180	"	"	"
CDR2-B-201D---	200	"	"	"
CDR2-B-221C---	220	"	"	200
CDR2-B-241C---	240	"	"	"
CDR2-B-271C---	270	"	"	"
CDR2-B-301C---	300	"	"	"
CDR2-B-331C---	330	"	"	"
CDR2-B-361C---	360	"	"	"
CDR2-B-391C---	390	"	"	"
CDR2-B-431C---	430	"	"	"
CDR2-B-471C---	470	"	"	"
CDR2-B-511B---	510	"	"	100
CDR2-B-561B---	560	"	"	"
CDR2-B-621B---	620	"	"	"
CDR2-B-681A---	680	"	"	50
CDR2-B-751A---	750	"	"	"
CDR2-B-821A---	820	"	"	"
CDR2-B-911A---	910	"	"	"
CDR2-B-102A---	1000	"	"	"
CDR2-B-112A---	1100	"	BP	"
CDR2-B-122A---	1200	"	"	"
CDR2-B-132A---	1300	"	"	"
CDR2-B-152A---	1500	"	"	"
CDR2-B-162A---	1600	"	"	"
CDR2-B-182A---	1800	"	"	"
CDR2-B-202A---	2000	"	"	"
CDR2-B-222A---	2200	"	"	"
CDR2-B-242A---	2400	"	"	"
CDR2-B-272A---	2700	"	"	"
CDR2-B-302A---	3000	"	"	"
CDR2-B-332A---	3300	"	"	"
CDR2-B-362A---	3600	"	"	"
CDR2-B-392A---	3900	"	"	"
CDR2-B-432A---	4300	"	"	"
CDR2-B-472A---	4700	"	"	"
CDR2-B-502A---	5000	"	"	"
CDR2-B-512A---	5100	"	"	"

1/ Complete type designation will include additional symbols to indicate style; voltage-temperature limits; capacitance tolerance (where applicable); termination finish ("T" for styles CDR21, CDR22, and CDR24, and "S" for styles CDR23 and CDR25), and failure rate level.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Multi-layer, unencapsulated, monolithic.
Material: Porcelain or ceramic.

Terminals:
Styles CDR21, CDR22, and CDR24: Silver leads attached with solder, with a melting temperature greater than 200°C.
Styles CDR23 and CDR25: No. 26 gage wire.

Inspection conditions: All visual examinations shall be performed using a stereo microscope having a minimum magnification of 20 power.

Capacitance: See table I.

Capacitance tolerance: See table I.

Rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Insulation resistance:
At +25° ±5°C: 10^6 megohms, minimum (for 0.1 pF thru 470 pF).
10^5 megohms, minimum (for 510 pF thru 1,000 pF).

At +125° ±5°C: 10^5 megohms, minimum (for 0.1 pF thru 470 pF).
10^4 megohms, minimum (for 510 pF thru 1,000 pF).

Quality factor: In accordance with MIL-C-55681.

Resonance: In accordance with MIL-C-55681.

Terminal strength: In accordance with MIL-C-55681.

Marking: In accordance with MIL-C-55681.

NOTE: These capacitors are intended for use in RF solid-state circuitry up to and including 25 gigahertz, and cannot be replaced in the field.

NOTE: Revision lettters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - ER
Navy - AS, OS
Air Force - 11, 17, 85
DLA - ES

User activities:
Navy - MC, SH
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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